

Exp No: 20

Code:

```
#include <lpc214x.h>

int main(void)
{
    unsigned sw_status;
    IODIR0 = 0x0000FF00;    // Configure Port0
    PINSEL0 = 0x00000000;    // Configure Port0 as General Purpose IO

    while(1)
    {
        sw_status = IOPIN0;
        IOSET0 = 0x0000FF00; // Set P0.15-P0.8 to '1'
        IOCLR0 = sw_status >> 8; // Set P0.15-P0.8 to '0'
    }
}
```

Output:

The screenshot displays the pVision IDE interface with the following components:

- Registers:** A list of registers (R0-R15, CPSR, SPSR) with their current values. R15 is highlighted with the value 0x00001D4.
- Disassembly:** A window showing the assembly code for the program. The instructions are:

```
9:          IOCLR0 = sw_sts >> 8; //Set P0.15-P0.8 to '0'  
0x000001D4 E1A01420 MOV     R1,R0,LSR #8  
0x000001D8 E582100C STR     R1,[R2,#0x000C]  
0x000001DC EAFFFFF7 B       0x000001C0
```
- Source Code (exp-20.c):** A window showing the C source code:

```
1 #include "lpc214x.h"  
2 int main(void){  
3     unsigned int sw_sts;  
4     IODIR0 = 0x0000FE00; //Configure Port0  
5     PINSEL0 = 0; //Configure Port0 as General Purpose IO  
6     while(1){  
7         sw_sts = IOPIN0;  
8         IOSET0 = 0x0000FF00; //Set P0.15-P0.8 to '1'  
9         IOCLR0 = sw_sts >> 8; //Set P0.15-P0.8 to '0'  
10    }  
11 }
```
- Command:** A window showing the command to load the program: "Load 'C:\\Users\\durga\\OneDrive\\Documents\\Objects\\exp-20.axf'". It also displays version information: "Restricted Version with 32768 Byte Code Size Limit" and "Currently used: 500 Bytes (1%)".
- Call Stack - Locals:** A window showing the call stack and local variables. The main function is listed with a location/value of 0x000001A4 and type int f(). The local variable sw_sts is listed with a location/value of 0x2FF00FF and type auto - unsigned int.
- Bottom Bar:** A status bar showing the simulation time (11: 4.25917992 sec), L9 C:1, and various system icons (temperature, weather, language, etc.).